

MC 302 DBMS: Relational Calculus

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Introduction

- Relational calculus is another formal query language used in the relational data model.
- Unlike relational algebra, which is procedural in nature and describes how to obtain the result, relational calculus is declarative and focuses on describing the desired result without specifying the method to achieve it.
- There are two main types of relational calculus:
 - Tuple Relational Calculus (TRC) and
 - Domain Relational Calculus (DRC).

Tuple Relational Calculus (TRC)

- In TRC, the query specifies what data should be retrieved based on certain conditions.
- The query is expressed as a set of conditions that the tuples in the result must satisfy.
- Variables are used to represent tuples, and the result is a set of tuples that satisfy the conditions.
- TRC is concerned with finding tuples that make a given formula true.

```
{ s.Name | Student(s) AND (∃ e)(Enrollment(e) AND e.StudentID = s.StudentID AND e.CourseName = 'Database Systems') }
```

TRC - Introduction

- a non-procedural query language used in relational databases to specify the retrieval of data without explicitly specifying how to obtain it.
- TRC is a declarative language, meaning that you describe what information you want to retrieve rather than how to retrieve it

$$\{t|P(t)\}$$

- Here, t is a tuple variable, and $P(t)$ is a predicate that defines the conditions for selecting tuples.
- Consider a relation *Students*(*StudentID*,*Name*,*Age*).
- A TRC query to find the names of students older than 20 years would be:

$$\{t|\text{Students}(t) \wedge t.\text{Age} > 20\}$$

- In words, this query retrieves tuples t from the *Students* relation where the *Age* attribute of t is greater than 20.

Key concepts in TRC

- **Tuple Variables:**

- $t, u, v, \dots$: These variables represent tuples from the specified relation(s).

- **Predicates:**

- $P(t)$: Predicates specify the conditions that tuples must satisfy. They are used to filter the tuples to be included in the result.

- **Logical Connectives:**

- Logical connectives such as $\wedge$ (AND), $\vee$ (OR), and $\neg$ (NOT) can be used to combine multiple predicates.

- **Quantifiers:**

- $\forall$ (for all): Specifies that the condition must be true for all tuples.
- $\exists$ (there exists): Specifies that there exists at least one tuple for which the condition is true.

TRC Example

$\{t | \text{Students}(t) \wedge \exists u (\text{Enrollment}(u) \wedge u.\text{StudentID} = t.\text{StudentID} \wedge u.\text{CourseName} = \text{'Database Systems'})\}$

- In words, this query retrieves tuples t from the Students relation for which there exists a tuple u in the Enrollment relation where $u.\text{StudentID}$ is equal to $t.\text{StudentID}$ and $u.\text{CourseName}$ is 'Database Systems'.

Domain Relational Calculus (DRC)

- In DRC, the query specifies what data should be retrieved based on certain conditions.
- The query is expressed as a set of conditions on the attributes (domains) of the relations.
- Variables represent values, and the result is a set of values that satisfy the conditions.
- DRC is concerned with finding values that make a given formula true.

```
{ Name |  $\exists$  StudentID, CourseName (Enrollment(StudentID, CourseName) AND Student(StudentID, Name)  
AND CourseName = 'Database Systems') }
```

DRC

- non-procedural query language used in relational databases to specify the retrieval of data without explicitly specifying how to obtain it.
- DRC is a declarative language, meaning that you describe what information you want to retrieve rather than how to retrieve it.

$$\{X|P(X)\}$$

- Here, X is a tuple variable, and $P(X)$ is a formula that defines the conditions for selecting tuples.
- Example: Consider a relation *Students(StudentID,Name,Age)*.
- A DRC query to find the names of students older than 20 years would be:

$$\{X|\text{Students}(X)\wedge X.\text{Age}>20\}$$

- this query retrieves tuples X from the *Students* relation where the *Age* attribute of X is greater than 20.

Key elements in DRC

- **Tuple Variables:**

- $X, Y, Z, \dots$: These variables represent tuples from the specified relation(s).

- **Formulas (Predicates):**

- $P(X)$: Formulas specify the conditions that tuples must satisfy. They are used to filter the tuples to be included in the result.

- **Quantifiers:**

- $\forall$ (for all): Specifies that the condition must be true for all tuples.
- $\exists$ (there exists): Specifies that there exists at least one tuple for which the condition is true.

- **Logical Connectives:**

- Logical connectives such as $\wedge$ (AND), $\vee$ (OR), and $\neg$ (NOT) can be used to combine multiple formulas.

DRC - Example

$\{X | \text{Students}(X) \wedge \exists Y (\text{Enrollment}(Y) \wedge Y.\text{StudentID} = X.\text{StudentID} \wedge Y.\text{CourseName} = \text{'DatabaseSystems'})\}$

- In words, this query retrieves tuples X from the Students relation for which there exists a tuple Y in the Enrollment relation where $Y.\text{StudentID}$ is equal to $X.\text{StudentID}$ and $Y.\text{CourseName}$ is 'Database Systems'.